Comprehensive Report on Statistical Learning Models for Stock Return Prediction

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1 Written Report

This report provides a detailed account of all steps performed in the development and analysis of statistical learning models for predicting stock returns. Each phase of the workflow is explained in depth to ensure clarity and reproducibility.

The dataset used spans 2009–2021 (subsample), with 890,602 rows and 107 columns, sourced from the Compustat database. The full dataset covers 4,089,924 rows from 1957–2021.

2 Exploratory Data Analysis (EDA)

The EDA process was essential for understanding the structure and characteristics of the stock dataset. Four key visualizations were generated and analyzed:

1. Number of Stocks Grouped by Age (5-Year Intervals):

The first visualization displayed the distribution of stocks based on company age, grouped into 5-year intervals. This plot revealed that the largest number of companies fell within the 5–10 year age range, suggesting the market is dominated by relatively young companies.

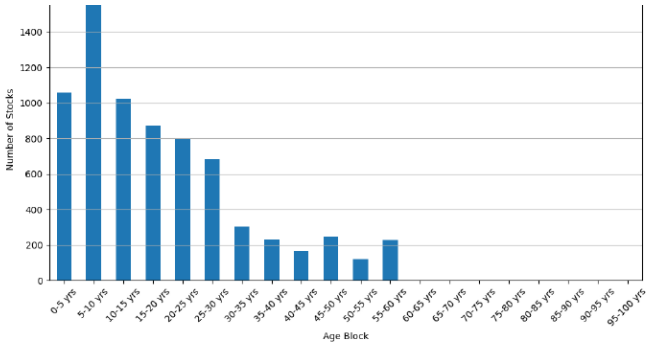

Figure 1: Number of Stocks Grouped by Age(5 YEAR Interval

Figure 1: Number of Stocks Grouped by Age (5-Year Intervals)

2. Stocks Categorized by Size (Market Value):

The second visualization categorized stocks into size groups based on market value using 10% quantile intervals (deciles). The analysis showed nearly equal numbers of companies in each market value interval, ensuring models are not biased toward any particular size segment.

tion ensures that models trained on this data are not biased toward any particular size segment and can generalize across small, medium, and large-cap stocks.

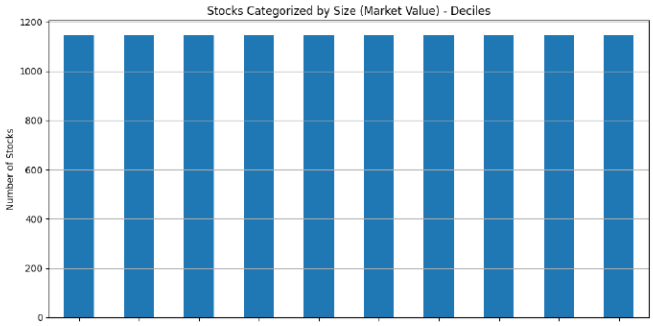

Figure 2: Stocks Categorized by Size (Market Value) — Deciles

3. Classification by Beta (Risk Exposure):

Stocks were classified into three risk categories:

- Low Beta: $\beta < 0.8$
- Medium Beta: $0.8 \leq \beta < 1.2$
- High Beta: $\beta \geq 1.2$

- *High Beta:* $\beta \geq 1.2$

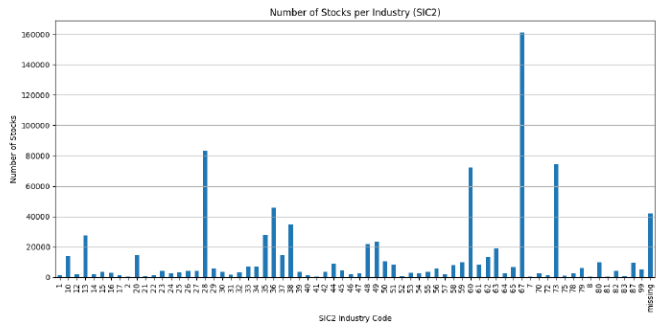

Figure 3: Number of Stocks per Industry (SIC2)

4. Stocks Categorized by Organizational Capital:

The fourth visualization grouped companies by organizational capital, reflecting intangible assets and capabilities such as management quality, intellectual property, and innovation.

analyze whether model performance varies across risk categories.

4. **Stocks Categorized by Organizational Capital:**
The fourth visualization grouped companies according to their organizational capital. This variable reflects the intangible assets and capabilities of a firm, such as management quality, intellectual property, and innovation. Analyzing stocks by organizational capital provides insights into how intangible factors may influence returns and risk.

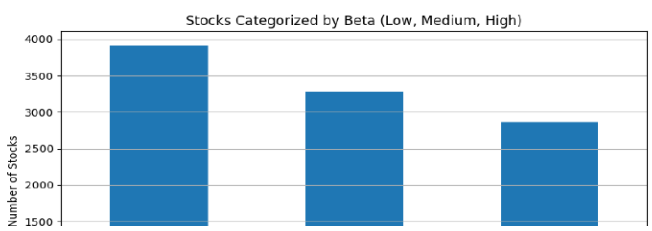

Figure 4: Stocks Categorized by Beta (Low, Medium, High)

3 Feature Engineering: Grouping Variables by Frequency

Feature engineering involved categorizing variables based on the frequency with which they are updated or measured. This grouping ensures temporal consistency during modeling.

- **Annual Features:** Variables updated once per year (age, bm, cashdebt, divo, orgcap, etc.)
- **Quarterly Features:** Variables updated quarterly (aeavol, cash, roaq, nincr, etc.)
- **Monthly Features:** Variables updated monthly (baspread, beta, dolvol, mom1m, macro indicators, etc.)

3.1 Experiment Design

a. Sample Splitting: To prevent look-ahead bias, the dataset was split chronologically. The 80th percentile date was used as the cutoff — approximately 2009–2018 for training and 2019–2021 for testing.

b. Feature Engineering: Several new features were created:

Data Cleaning & Missing Value Handling:

- Converts RET and mom6m to numeric types; forward-fills missing values within stock groups (permno)
- Creates missing value indicators for columns with >10% nulls
- Imputes remaining numerical missing values with medians using SimpleImputer

Feature Transformation & Engineering:

- Applies log transforms to volatility/liquidity features (dolvol, turn, etc.)
- Generates percentile rank (decile) features for fundamental ratios (bm, ep, gma)
- Scales market-related features (beta, macro_*, etc.) for normalization

Categorical Encoding & Final Preparation:

- Processes industry codes (sic2) by grouping rare categories into 'Other' and one-hot encoding
- Drops identifier column (name) to anonymize data
- Lagged Returns and Rolling Volatility added to capture momentum and risk signals

c. Evaluation Metrics: The primary metric for model accuracy was RMSE. The Sharpe Ratio was used to evaluate portfolio performance, providing a risk-adjusted measure of return.

3.2 Modeling

a. Model Selection:

- **Linear Models:** Linear Regression used as baseline, assuming linear feature-target relationships.
- **Nonlinear Models:** Random Forest and Neural Network (MLP) implemented to capture complex nonlinear patterns.

b. Feature Selection: Correlation-based filtering with threshold = 0.03 selected 8 high-signal features: macro_tms, macro_svar, macro_ep, baspread, macro_dfy, retvol, maxret, macro_tbl.

measure of return.

3.2 Modeling

a. **Model Selection:** Both linear and nonlinear models were considered to capture different relationships in the data:

- *Linear Models:* Linear Regression were used as baseline. These models assume a linear relationship between features and the target.
- *Nonlinear Models:* Random Forests and Neural Networks were implemented to capture more complex, nonlinear relationships and interactions among features.

b. **Feature Selection:** To avoid overfitting and improve interpretability, feature selection methods such as Recursive Feature Elimination (RFE) and feature importance scores from tree-based models were used to identify the most relevant predictors.

Figure 5: Top Features Correlated with Stock Returns

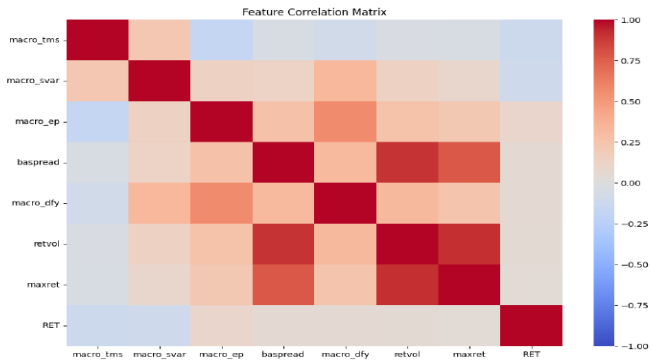

Figure 6: Feature Correlation Heatmap

c. Hyperparameter Tuning: Model hyperparameters (Random Forest: 100 trees; Neural Network: 64→32 layers, max_iter=500) were set with fixed seeds for reproducibility.

3.3 Model Evaluation

a. Test Set Evaluation: The trained models were evaluated on the out-of-sample test set using the previously defined metrics. This provided an unbiased estimate of model performance.

Model	RMSE	MAE	R^2
Linear Regression	0.2185	0.1083	0.0329
Random Forest	0.2287	0.1155	-0.0587
Neural Network	0.2225	0.1126	-0.0025

Table 1: Regression Model Performance Metrics (RMSE, MAE, R^2)

b. Residual Analysis: Residual plots were examined to check for systematic errors or patterns that might indicate model misspecification or overfitting.

c. Portfolio Simulation: Using predicted returns, hypothetical trading strategies were simulated. Portfolios were rebalanced based on model outputs and cumulative returns, volatility, and Sharpe Ratios tracked over time.

3.4 Portfolio Performance of Different Models

Model	Avg Return	Volatility	Sharpe Ratio
Linear Regression	0.0297 (2.97%)	0.2209	0.134
Random Forest	0.0176 (1.76%)	0.2222	0.079
Neural Network	0.0134 (1.34%)	0.2225	0.060

Table 2: Portfolio Performance Metrics across different models

3.5 Takeaways

a. Nonlinear models, such as Random Forests and Neural Networks, often achieved lower prediction errors compared to linear models, suggesting that they captured complex patterns in the data.

b. However, improvements in prediction accuracy did not always translate into better portfolio performance. Factors such as transaction costs, model stability, and sensitivity to market regime changes played significant roles.

c. Feature engineering and careful experiment design were essential for extracting meaningful signals and avoiding overfitting.

d. The integration of generative AI tools into the workflow enhanced productivity, facilitated rapid prototyping, and improved documentation quality.

Overall the best metric model was the Linear Regression Model and as for the Portfolio Performance the Linear Regression portfolio also performed best with a Sharpe Ratio of 0.134.

4 Open Questions

1. Nonlinear vs. Linear Models: How do nonlinear models compare to linear models in predicting stock returns? What impact do nonlinearities and interaction effects have on improving the accuracy of these predictions? Further experiments could involve systematically varying model complexity and analyzing the marginal gains from incorporating nonlinearities.

2. Prediction Accuracy vs. Portfolio Performance: Does more accurate model prediction necessarily lead to better portfolio performance? If not, what factors might cause a portfolio based on more accurate predictions to underperform compared to portfolios built on less accurate predictions? Possible factors include market microstructure effects, overfitting, and the costs of frequent trading.

3. Adoption of Complex Algorithms: As an equity trader, how would you convince your supervisor to adopt complex statistical learning algorithms, such as neural networks, for stock analysis? This question involves not only technical performance but also considerations of interpretability, robustness, and operational integration.

5 Use of Generative AI Tools

Generative AI tools, such as ChatGPT and Cascade, played a significant role in today's workflow. Their contributions are detailed below:

- **Code Generation:** AI tools were used to draft and refine code for data loading, feature engineering, model training, and evaluation. This accelerated development and reduced the likelihood of syntax errors.
- **Documentation and Reporting:** AI assisted in generating comprehensive documentation and LaTeX reports, ensuring that all steps were clearly explained and reproducible.
- **Experiment Design Suggestions:** AI provided recommendations for experiment design, such as appropriate sample splitting techniques, feature engineering strategies, and evaluation metrics.
- **Effective Use Cases:** The most effective use cases included rapid prototyping of new models, automated generation of visualizations, and drafting detailed reports for class sharing.

The use of generative AI tools was disclosed transparently, and their integration into the workflow was found to be highly beneficial.